

Workflow		Rank				HGSO
All		All				All
sequential	1	0.926	0.907	0.856	0.943	0.943
hybrid_xgb	2	0.935	0.922	0.802	0.899	0.791
up_mr	3	0.93	0.917	0.817	0.881	0.844
smote_mr	4	0.932	0.912	0.813	0.889	0.844
smote_xgb	5	0.938	0.911	0.797	0.892	0.828
two_step	6	0.926	0.917	0.814	0.875	0.872
hybrid_mr	7	0.928	0.911	0.821	0.872	0.843
up_xgb	8	0.924	0.9	0.803	0.882	0.696
hybrid_svm	9	0.923	0.893	0.832	0.86	0.814
smote_svm	10	0.923	0.875	0.826	0.869	0.814
down_mr	11	0.906	0.909	0.776	0.87	0.874
hybrid_rf	12	0.923	0.898	0.806	0.894	0.689
up_svm	13	0.91	0.862	0.799	0.851	0.759
up_rf	14	0.905	0.894	0.808	0.878	0.59
smote_rf	15	0.911	0.872	0.797	0.865	0.595
down_rf	16	0.892	0.905	0.746	0.815	0.877
down_xgb	17	0.883	0.91	0.753	0.822	0.789
none_svm	18	0.882	0.879	0.778	0.868	0.5
none_rf	19	0.879	0.895	0.741	0.858	0.583
down_svm	20	0.879	0.873	0.757	0.906	0.906
none_mr	21	0.5	0.5	0.5	0.5	0.5
none_xgb	22	0.505	0.5	0.499	0.5	0.5